Zero-Shot Multi-Modal Part Segmentation for Space Target via 3D-2D Semantic Priors Transfer

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***Abstract*—Accurate satellite component segmentation is critical for Space Situational Awareness (SSA), enabling essential operations such as on-orbit servicing, anomaly diagnosis, and autonomous missions. Current 2D-based approaches face significant limitations. For example, supervised methods require extensive annotated data that is impractical for space applications, and few-shot learning suffers from poor generalization and computational complexity. Meanwhile, foundation models like SAM produce class-agnostic masks that lack semantic awareness to distinguish critical satellite components. To overcome these 2D-only limitations, we propose a novel multi-modal zero-shot learning framework that leverages 3D satellite information. Our approach transfers semantic priors from 3D point cloud data to 2D imagery, enabling simultaneous component segmentation across both modalities without requiring annotated training data. Specifically, we first enhance 3D zero-shot semantic cues through spatial-guided 3D semantic smoothing and geometry-aware antenna isolation. These refined 3D priors are then integrated into a prompt-automatic consensus 2D segmentation module, empowering it with explicit semantic guidance. Furthermore, to support cross-modal evaluation, we introduce a comprehensive multi-modal satellite dataset NASA-2/3D-Part comprising mesh models, point clouds, multi-view synthetic images, and ground truth segmentation masks, establishing a standardized benchmark for satellite component segmentation methods. Extensive experiments demonstrate superior performance compared to existing methods, showing significant improvements in semantic accuracy and generalization capability for space missions.**

***Index Terms*—Satellite part segmentation, zero-shot learning, multi-modal learning, foundation model, point cloud**

I. Introduction

With the continuous advancement of space exploration, precise perception of space targets has become critical for Space Situational Awareness (SSA). Accurate part segmentation of satellite components, including the main bus, solar arrays, and antennas, is essential for on-orbit operations, anomaly diagnosis, and autonomous servicing. Concurrently, deep learning has experienced rapid development in recent years and has been increasingly integrated into SSA tasks [1–4]. These deep learning-driven approaches have demonstrated superior performance in handling complex space environments, multi-modal data fusion, and real-time processing requirements that are fundamental to modern SSA systems.

Current satellite component segmentation methodologies predominantly focus on the 2D modality [5–10]. Among these approaches, traditional supervised learning-based methods [8, 9] require extensive annotated 2D data, which is time-consuming and labor-intensive, making them impractical for space applications. To address this, Few-Shot Learning (FSL) [3, 4] reduces data dependency by leveraging limited samples. However, FSL methods struggle with insufficient cross-category generalization and computationally intensive meta-learning pipelines, limiting their scalability. These challenges have led to the exploration of Zero-Shot Learning (ZSL) [11, 12], which eliminates the need for annotated training data entirely. While promising, existing ZSL methods are designed for general-purpose tasks and fail to address the unique complexities of satellite components and their configurations.

Recent advances in foundation models, such as the Segment Anything Model (SAM) series [13–16], have demonstrated remarkable capabilities in zero-shot 2D image segmentation tasks. However, these models fundamentally output class-agnostic masks, which lack the semantic awareness necessary to distinguish critical satellite components like solar panels, antennas, and main bus structures. This limitation restricts their direct applicability to space missions. Moreover, SAM's reliance on manual prompts makes the process labor-intensive and impractical, while its automatic prompt generation is highly noise-sensitive. This often results in issues such as redundant masks or missing critical components.

To address the aforementioned challenges, we propose a novel multi-modal framework that leverages 3D satellite information to overcome the inherent limitations of 2D-only approaches. Our method transfers zero-shot semantic priors from 3D point cloud data to 2D space, enabling simultaneous component segmentation across both modalities without requiring annotated training data. Specifically, we enhance the precision of 3D zero-shot semantic cues through smoothing vision-language model (VLM) [17] features of each 3D point and implementing geometry-aware antenna isolation that exploits the distinctive convex hull characteristics. These refined 3D semantic priors are then utilized to empower the SAM model by providing explicit semantic guidance to its class-agnostic masks. Finally, we integrate a prompt-automatic consensus 2D segmentation module with a dual-stage fusion mechanism to ensure robust 2D segmentation results.

Furthermore, recognizing the absence of publicly available multi-modal satellite datasets for such applications [8–10, 18–20], we introduce a comprehensive dataset, NASA-2/3D-Part, comprising 109 satellite mesh models, point clouds, multi-view synthetic images, and their corresponding ground truth segmentation masks for critical components. The acquisition and statistics will be elaborated on in a later section. Bene-

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fiting from the multi-modal dataset, our method involves two branches for processing images and point clouds. The dataset provides a standardized benchmark for evaluating multi-modal satellite component segmentation methods.

Our contributions can be summarized as follows:

- A novel multi-modal zero-shot learning framework that bridges 3D point cloud semantic priors with 2D image segmentation for satellite components;
- A comprehensive multi-modal satellite dataset with precise component-level annotations;
- Extensive experiments demonstrate the superior performance of our ZSL method for space missions.

II. Related Work

A. Satellite Component Segmentation

Satellite Component Segmentation refers to the task of identifying and delineating distinct functional parts of satellites, such as the main body, solar panels, antennas, and other appendages. The research on this task primarily focuses on 2D modality. Early research [21, 22] relied on manually crafted features and hand-tuned parameters, which demonstrated limited adaptability and generalization across diverse satellite configurations. Contemporary approaches [5–10] leverage deep learning architectures and achieve substantially superior performance in terms of accuracy and robustness. However, these methods invariably require extensive manually annotated images for training, making them impractical for real-world deployment. Zhao et al. [23] extended the task to the 3D domain by introducing point cloud processing networks for space targets, yet this approach remains constrained within the traditional supervised learning paradigm. In this paper, we propose a novel multi-modal framework that effectively bridges 3D point cloud semantic priors with 2D image segmentation for satellite components. Moreover, our approach employs zero-shot learning for both 3D and 2D data processing, thereby eliminating the dependency on expensive annotated data and computationally intensive training processes.

B. Zero-Shot Learning

Zero-Shot Learning (ZSL) aims to classify or segment unseen categories without requiring labeled training data. Prior to the advent of foundation models, zero-shot segmentation tasks [24, 25] were typically addressed by integrating visual features extracted from traditional convolutional neural networks (CNNs [26]) with semantic embeddings derived from textual descriptions or word vectors. However, they often struggle with the inherent challenges posed by long-tail label distributions—a problem that is particularly pronounced in satellite image segmentation, where fine-grained semantic components typically fall within the tail of the data distribution, making them difficult to segment accurately. Foundation models, such as the Segment Anything Model (SAM) series [13], have demonstrated impressive capabilities in generating class-agnostic masks, but these masks often lack the semantic detail necessary to distinguish critical satellite components, such as solar panels and antennas. Furthermore, SAM's reliance on manual prompts is labor-intensive, while its automatic prompt generation is prone to noise, resulting in redundant or missing masks. In this paper, we propose a novel approach that transfers zero-shot semantic priors from 3D point cloud data to 2D space, providing explicit semantic guidance for class-agnostic mask predictions. Additionally, we introduce a dual-mode SAM segmentation module equipped with a hierarchical voting fusion mechanism to ensure more robust and accurate 2D segmentation results.

C. Multi-Modal Learning

Multi-modal learning frameworks harness complementary information across diverse data modalities to achieve superior model performance and enhanced robustness. Extensive research efforts [27–31], exemplified by approaches like MV3D [30] and PVRNet [28], have validated the efficacy of exploiting cross-modal representations. However, satellite component segmentation methodologies predominantly focus on the 2D modality and supervised learning paradigm[5–10]. Building upon these insights, we introduce a novel multi-modal learning approach that combines heterogeneous information sources to enhance the accuracy of satellite component segmentation.

D. Space Domain Datasets and Benchmarks

The inherent challenges associated with acquiring large-scale real-world space target datasets have established synthetic datasets as the predominant approach for developing deep learning methodologies. SPEED [20] and SwissCube [32] are specifically designed for 6D pose estimation tasks. Satellite Component Video Dataset [10] encompasses 13 satellite models with corresponding 2D imagery, making it suited for evaluating our multi-modal framework. Unfortunately, this dataset remains unavailable for public access. Given the zero-shot nature of our approach and the substantial scale of our proposed dataset containing 109 distinct models, the independent evaluation across all 109 models provides compelling evidence of our method's generalization capabilities.

III. Methodology

A. Overview

Our proposed framework aims to achieve zero-shot semantic segmentation of satellite components across both 2D imagery and 3D point clouds. The overall pipeline is depicted in Fig. 1, takes as in put a clean 3D point cloud P, its corresponding mesh model M, and a set of multi-view 2D images $\{I_m, K_m, E_m\}_{m=1}^{K}$.

Initially, the Spatially-guided 3D Semantic Smoothing module extracts local geometric features, including surface normals n_i and FPFH f_i directly from the input point cloud to form geometrically and semantically coherent 3D primitives. Subsequently, a Geometry-aware Antenna Isolation algorithm is employed. This algorithm leverages graph segmentation and normal vector convergence analysis to identify and extract antenna structures, which exhibit distinctive high-curvature and directional surface properties. These pre-segmented antenna points P_a are preserved for later reintegration into the

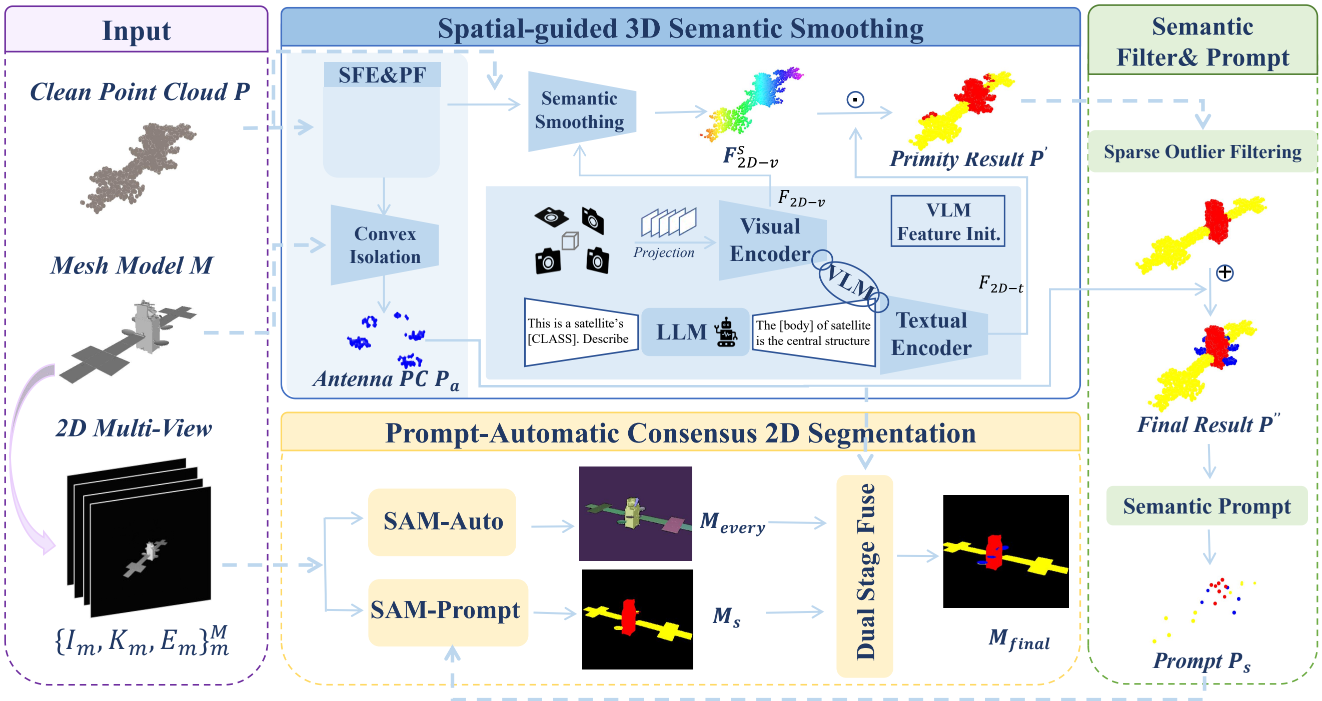

Fig. 1. Overall pipeline of 2D view and 3D PC segmentation. *Left:* input of our model, including clean point cloud P, mesh model M, and according 2D images; *Top:* it shows the spatial-guided VLM denoising process. This includes the GPT-4 model that generates detailed descriptions of images, the process of extracting and aggregating geometric features of point cloud data and VLM features, and antenna isolation preprocessing; *Right:* fuse the segmentation point cloud primary results P' and the antenna preprocessing point cloud results, then input them into clustering to select prompt points.*Bottom:* the SAM module performs automatic segmentation and point cloud prompt point input segmentation, and fuses and corrects the output 2D view segmentation results.

final segmentation. In parallel, GPT-4 and VLM are utilized to generate text features F_{2d-t} and visual features F_{2d-v} which are projected onto the 3D point cloud to obtain initial semantic features.These initial VLM features are then refined through a semantic smoothing process. The point cloud is first partitioned into superpoints by optimizing spatial, geometric, and semantic coherence. A hierarchical aggregation is then performed, involving local smoothing within each superpoint and global propagation across structurally similar regions, resulting in denoised and spatially consistent semantic features F^s_{2d-v}. An initial 3D segmentation P' is obtained by assigning each point the semantic label whose text feature F_{2d-t} has the highest similarity to its smoothed visual feature F^s_{2d-v}.

The initial segmentation P' and the pre-segmented antenna points P_a are then fed into the Semantic Filtering and Prompt Generation module. Here, a two-stage filter first refines the segmentation to produce P'', from which category-specific 3D prompt points P_s are extracted via adaptive clustering.

These 3D prompt points P_s are projected onto each 2D image plane to serve as explicit semantic prompts for the subsequent 2D segmentation module. This module employs a dual-mode segmentation strategy using SAM. In the prompt mode, the projected points guide the generation of semantically informed masks M_{body} and M_{wing}. Concurrently, in the automatic mode, SAM produces a comprehensive set of class-agnostic masks M_{every} that densely partition the image. Finally, the Dual Stage Fusion mechanism integrates these two sets of masks to obtain M_{final}.

B. Spatial-guided 3D Semantic Smoothing

Spatial Feature Extraction & Primitive Formation. We extract point-level spatial descriptors and aggregate them into geometric primitives for later convex isolation and semantic smoothing.

Given the clean point cloud $P = \{\boldsymbol{p}_c \in \mathbb{R}^3\}$, for each $\boldsymbol{p}_c$ we estimate its vertex normal $\mathbf{n}_i$ and compute its Fast Point Feature Histograms (FPFH[33]) features $\boldsymbol{f}_i \in \mathbb{R}^d$ using Open3D library[34].

Let the vertex set of the Mesh model be $V = \{\mathbf{v}_i \in \mathbb{R}^3 \mid i = 1, \ldots, n\}$ and the face set be $F = \{f_j\}$. First, we compute the smoothed normal vector $\mathbf{n}_j$ for each vertex:

$$\mathbf{n}_{f_i} = \frac{(\mathbf{v}_\beta - \mathbf{v}_\alpha) \times (\mathbf{v}_\gamma - \mathbf{v}_\beta)}{\|(\mathbf{v}_\beta - \mathbf{v}_\alpha) \times (\mathbf{v}_\gamma - \mathbf{v}_\beta)\|}, \tag{1}$$

$$\mathbf{n}_j = \frac{1}{|N(j)|} \sum_{f_i \in N(j)} \mathbf{n}_{f_i}, \tag{2}$$

$N(j)$ is the set of all faces adjacent to vertex $\mathbf{v}_j$.

A graph $G = (V, E)$ is constructed based on the mesh structure. The edge set E is induced by the face set F, containing all edges of the triangular faces: $E = \{(\mathbf{v}_i, \mathbf{v}_k) \mid \exists f_j \in F \text{ s.t. } \{\mathbf{v}_i, \mathbf{v}_k\} \subset f_j\}$. The weight $w(e_{ij})$ of an edge connecting vertices $\mathbf{v}_i$ and $\mathbf{v}_j$ is primarily defined by the difference between their normal vectors. A convexity adjustment is applied by squaring this base weight if the edge represents a convex transition. The final weight is determined as follows:

$$w(e_{ij}) = \begin{cases} (1 - |\mathbf{n}_i \cdot \mathbf{n}_j|)^2 & \text{if } d_{ij} > 0 \\ 1 - |\mathbf{n}_i \cdot \mathbf{n}_j| & \text{otherwise} \end{cases}, \tag{3}$$

where the convexity indicator d_{ij} is the dot product of the normal vector $\mathbf{n}_j$ and the normalized direction vector from $\mathbf{v}_i$ to $\mathbf{v}_j$.

Subsequently, the graph is segmented by an iterative merging process. Edges are considered in ascending order of their weight w. A merge between two distinct components C_a and C_b is performed if the connecting edge's weight w does not exceed their respective internal thresholds, $\tau(C_a)$ and $\tau(C_b)$. Following a merge, the threshold for the newly formed component C_{new} is updated using the rule $\tau(C_{\text{new}}) = w + c/|C_{\text{new}}|$, where c is a parameter that controls the preference for larger components.

This iterative merging process yields a set of segmented clusters S_k, each representing a spatially coherent geometric primitive for subsequent antenna detection.

Convex Isolation. Having obtained the geometric primitives S_k from the graph segmentation stage, we now perform antenna detection through normal vector convergence analysis. Antenna structures on mesh models, such as parabolic dishes and horn antennas, are characterized by high curvature and strong directional alignment of surface normals toward a focal point. This geometric property enables precise segmentation of antennas from the satellite body, with extracted antenna regions reserved for later reintegration.

To quantify the likelihood of each cluster S_k being an antenna, we define an antenna score based on two complementary metrics: the convergence of normal vectors (measuring directional alignment) and their variance (penalizing excessive angular dispersion). Clusters exceeding a predefined score threshold are classified as antenna candidates. Subsequently, an adaptive matching algorithm is employed to accurately transfer these mesh-based antenna labels to the corresponding 3D point cloud.

For each segmented cluster S_k, We first compute the following intermediate quantities:

$$\bar{\mathbf{n}}_k = \frac{1}{|S_k|} \sum_{\mathbf{n}_i \in S_k} \mathbf{n}_i. \tag{4}$$

$$\mathrm{CS}_k = \frac{1}{|S_k|} \sum_{\mathbf{n}_i \in S_k} \max(0,\ \mathbf{n}_i \cdot \bar{\mathbf{n}}_k). \tag{5}$$

$$\mathrm{VS}_k = \frac{1}{|S_k|} \sum_{\mathbf{n}_i \in S_k} \left\| \mathbf{n}_i - \bar{\mathbf{n}}_k \right\|^2. \tag{6}$$

Here, CS_k measures directional alignment (higher is better), whereas VS_k denotes the within-cluster normal variance and penalizes angular dispersion (lower is better).

Given these intermediate quantities, we finally compute the antenna score:

$$\mathrm{AS}_k = \delta_1 \cdot \mathrm{CS}_k \ + \ \delta_2 \cdot \left(1 - \frac{\mathrm{VS}_k}{\delta_3}\right), \tag{7}$$

where δ_1 and δ_2 weight the alignment and dispersion terms, and δ_3 normalizes the variance. A segment is classified as an antenna if AS_k exceeds a predefined threshold δ.

Since the point cloud is generated by sampling the faces of the original mesh model, the antenna labels identified on the mesh must be accurately transferred to their corresponding points. To this end, we design an adaptive matching algorithm to achieve precise alignment between the point cloud and the identified antenna regions from the mesh model.We update point cloud labels through nearest neighbor search and an adaptive thresholding criterion.

Specifically, the reference set of antenna points be $R = \{\mathbf{r}_i \mid i = 1, \ldots, m\}$, where $\mathbf{r}_i$ are the coordinates of points labeled as antenna in the Mesh model. Let the point cloud set be $P = \{\mathbf{p}_c\}$.

For each point $\mathbf{p}_c$ in the point cloud, we compute its minimum Euclidean distance to the reference set R:

$$d_a = \min_{i=1,\ldots,m} \|\mathbf{p}_a - \mathbf{r}_i\|_2. \tag{8}$$

We then establish an adaptive threshold, τ_a, to determine the label assignment. Specifically, if the computed distance d_a is less than this threshold ($d_a < \tau_a$), an antenna' label is assigned to $\mathbf{p}_c$. This generates the segmented antenna point cloud $P_{\text{antenna}} = \{(\boldsymbol{p}_a, \text{label}(\boldsymbol{p}_a))\}_{a=1}^{N}$. The threshold τ_a is adaptive, dynamically adjusted based on local point cloud density.

$$\tau_a = \rho \cdot f, \tag{9}$$

$$\rho = \frac{1}{n'} \sum_{c=1}^{n'} \min_{j \neq c} \|\mathbf{p}_c - \mathbf{p}_j\|_2. \tag{10}$$

where ρ represents the average nearest neighbor distance of the point cloud, which characterizes its density, and f is a scaling factor. If matching fails, the algorithm switches to the larger threshold.

This adaptive mechanism ensures the algorithm's robustness across different satellite models, exhibiting superior performance, especially in point clouds with significant noise or scale variations.

VLM Feature Initialization. This module aims to endow the 3D point cloud with semantic features by harnessing knowledge from 2D VLMs. Specifically, GPT-4 [35] and CLIP [36] are employed to generate textual and visual features, respectively, which are then associated with each 3D point. A unified prompt template is utilized for all satellite component classes: "This is a satellite's [CLASS]. Please describe its detailed appearance for segmentation, including the outline boundaries, shape, relative position, and any distinctive structural features."

For each class $n \in \{\text{body}, \text{antenna}, \text{wing}\}$, GPT-4 is queried ten times to obtain a set of diverse text descriptions T_n, which is defined as

$$T_n = GPT - 4(Prompt). \tag{11}$$

Subsequently, the T_n are fed into CLIP's text encoder to obtain the corresponding textual features $\boldsymbol{F}_{2D-t}$. This process is defined as:

$$\boldsymbol{F}_{2D-t} = \mathrm{CLIP}_{\text{text}}(T_n) \tag{12}$$

To extract visual features by following the common practice in [37, 38], we project the point cloud onto multiple predefined viewpoints to generate image set I_v. These images are then processed by CLIP's visual encoder to produce dense visual features $\boldsymbol{F}_{2D-v}$ defined as:

$$\boldsymbol{F}_{2D-v} = \mathrm{CLIP}_{\mathrm{visual}}(I_v) \tag{13}$$

Finally, both types of 2D features are projected onto the 3D point cloud, resulting in the initial 2D VLM feature representation for each 3D point.

Semantic Smoothing.

Direct projection from 2D images to 3D point clouds often introduces local semantic noise, caused by view occlusion and projection inaccuracies. Additionally, the inherent geometric correlations of 3D point clouds are not considered. Consequently, spatially adjacent points with similar geometry may be assigned inconsistent semantic features, undermining the overall semantic coherence. To address this, we introduce a semantic smoothing mechanism that leverages geometric features to denoise the initial VLM features and enforce semantic consistency.

Our semantic smoothing pipeline consists of the following steps. First, the point cloud P is partitioned into geometrically and semantically coherent superpoints S_j. This is achieved by optimizing a cost function that balances spatial proximity, geometric consistency using point normals n_i and FPFH features f_i, and semantic affinity derived from the initial features F_{2d-v}. Then, the features are optimized in two steps through a hierarchical aggregation process. The first step is to perform local smoothing within the superpoint neighborhood $\mathcal{N}_j$, fusing the embedded features of points with similar geometry and semantics to eliminate local noise. The second step is to conduct global propagation, allowing structurally similar superpoints to exchange contextual information. This hierarchical aggregation ensures global semantic consistency. The final smoothed visual feature for a point p_i, denoted $\mathbf{F}^s_{2d-v}(p_i)$, is the output of this process. The local smoothing for a point within superpoint $\mathcal{S}_j$ can be formulated as:

$$\mathbf{F}_{\mathrm{local}}(p_i) = \frac{1}{Z} \sum_{p_k \in \mathcal{N}(p_i)} w_{ik} \cdot \mathbf{F}_{2d-v}(p_k), \tag{14}$$

where w_{ik} is a weight based on geometric and semantic similarity, and Z is a normalization factor. The global propagation refines the superpoint-level features $\mathbf{F}_{\mathrm{local}}(\mathcal{S}_j)$ via graph convolution.

Finally, the initial segmentation result P' is obtained by calculating the dot product of the smoothed VLM visual features and text features:

$$P' = F^s_{2d-v} \cdot \boldsymbol{F}_{2d-t} \tag{15}$$

C. Semantic Filter And Prompt.

In this section, we apply a semantic filter on the initial segmentation result P' and the pre-segmented antenna result P_a to obtain the final segmentation result P''. Then the prompt points P_s are selected via semantic prompt algorithm.

Sparse Outlier Filtering. The fine-grained and long-tailed nature of the antenna category introduces significant noise into the VLM-based semantic understanding, which consequently adversely affects the segmentation of other categories. To address this issue, we employ a two-stage filtering approach to refine the initial segmentation P'.

Stage 1: Due to the unreliability of VLM-derived antenna predictions, we first remove all points labeled as antennas from the initial segmentation P'. Subsequently, to mitigate the noise in the remaining antenna predictions, a majority-voting filter is applied within the local neighborhood of each point originally labeled as antenna in P'. Specifically, for each such point, its k-nearest neighbors in 3D space are identified, and its label is replaced by the majority label within this neighborhood, effectively suppressing spurious antenna predictions. The pre-segmented antenna region P_a is then overlaid onto the processed output, resulting in a refined antenna segmentation devoid of antenna-related semantic noise interference.

Stage 2: To address outliers in the non-antenna categories from the initial segmentation P', an outlier removal strategy is implemented. For each non-antenna point, the majority label among its k-nearest neighbors is computed. If the point's original label differs from this majority label, it is updated to the majority label to align with the local consensus, thereby eliminating segmentation outliers. The result of this two-stage filtering is the final refined 3D segmentation P''.

Semantic Prompt. To generate prompt points suitable for SAM, we extract representative points from the refined segmentation P''. Specifically, adaptive clustering is performed with the maximum neighborhood radius parameter $epsilon$ set as 1.5× median of k-nearest neighbor distances. For each resulting cluster $\mathcal{S}_k$, where $k = 1, 2, \ldots, K$, a single semantic prompt point is generated[39]. The centroid of each cluster is computed as:

$$\mathbf{p}_k = \frac{1}{|\mathcal{S}_k|} \sum_{p_n \in \mathcal{S}_k} p_n, \tag{16}$$

where $|\mathcal{S}_k|$ denotes the number of points in the cluster, and P_n represents the 3D coordinates of a point.

To assign semantic labels to these centroids, we utilize the previously generated segmentation resul labels. Specifically, for each centroid $\mathbf{p}_k$, we identify its nearest neighbor in the final segmentation P'' and assign the corresponding semantic label s from the set $\{\mathrm{body}, \mathrm{antenna}, \mathrm{wing}\}$, thereby obtaining the labeled prompt points $\{P_s\}$.

D. Prompt-Automatic Consensus 2D Segmentation

This module employs a hybrid SAM segmentation approach with a dual fusion module. It strategically combines auto mode for superior boundary handling with prompt mode for semantic guidance.

SAM-Prompt. The categorized prompt points P_s and 2D views I_m are input into the segmentation model $\mathrm{SAM}_{\mathrm{prompt}}$ to produce semantically-aware segmentation results.

The projection of the k-th prompt point $P_{s,k}$ in category s onto the m-th image is achieved using the standard pinhole camera model:

$$(P^m_{s,k}, 1)^T = K_m \cdot E_m \cdot (P_{s,k}, 1)^T, \tag{17}$$

where K_m denotes the camera intrinsic matrix, E_m represents the camera extrinsic matrix, and $P^m_{s,k}$ is the projected 2D point.

The projected points $\{P^m_{s,k}\}$ and image I_m are fed into $\mathrm{SAM}_{\mathrm{prompt}}$. Segmentation mask candidates are predicted as:

$$M_s = \mathrm{SAM}_{\mathrm{prompt}}(I_m, \{P^m_{s,k}\}_{k=1}^{K_s}), \tag{18}$$

where K_s is the number of prompt points for category s.

Separate segmentation is performed for body and wings, resulting in M_{body} and M_{wing} respectively.

SAM-Auto. The SAM is employed in its automatic mode to process the input 2D image I_m In this mode, the model autonomously generates an initial set of candidate points as its primary step. These points serve as implicit prompts, and the model further produces a collection of masks M_{every} that partition the image into numerous distinct regions, which is defined as:

$$M_{\mathrm{every}} = \mathrm{SAM}_{\mathrm{auto}}(I_m), \tag{19}$$

where $\mathrm{SAM}_{\mathrm{auto}}$ represents the SAM model operating in automatic segmentation mode. This mode eliminates the need for manual prompts and directly outputs a diverse range of masks. These masks cover various objects and regions within the input 2D view I_m. Each mask within M_{every} corresponds to a specific segmented block, and all masks collectively decompose the original image into a set of non-overlapping regions. Note that partial overlaps may exist only in the initial mask generation stage, and they are removed via post-processing.

Dual Stage Fusion. The masks generated by SAM-Auto and SAM-Prompt are fused through a dual-stage process to produce the final segmentation result.

The automatically generated masks in M_{every} often contain overlapping regions. Specifically, for all masks with size smaller than β, they are processed in descending order of size, with smaller masks overriding the class definitions of larger ones. For each such small mask $M_{\mathrm{smaller}} \in M_{\mathrm{every}}$ (with size < β), its category is determined by calculating the overlap with the prompt-based masks. Specifically, the category is assigned to the one with higher overlap:

$$O_b = |M_{\mathrm{smaller}} \cap M_{\mathrm{body}}|, \quad O_w = |M_{\mathrm{smaller}} \cap M_{\mathrm{wing}}|. \tag{20}$$

$$\mathrm{Category}(M_{\mathrm{smaller}}) = \begin{cases} \text{body}, & O_b \geq O_w \\ \text{wing}. & \text{otherwise} \end{cases} \tag{21}$$

M_{fused} is constructed by integrating all categorized masks from M_{s} and supplementing regions from the prompt-based masks M_{body} and M_{wing} to cover any unassigned areas.

Based on the antenna segmentation result P_{a}, the 3D antenna point cloud is projected onto the corresponding 2D view using the orthographic projection method defined in equation(17), generating a binary antenna mask M_{antenna} that accurately delineates antenna regions in the image space for subsequent segmentation integration.

The final mask integrates the antenna regions through a dedicated replacement layer, completing the dual stage fusion process that combines body, wing, and antenna information into a segmentation result M_{final} which is defined as:

$$M_{\mathrm{final}} = (M_{\mathrm{fused}} \setminus M_{\mathrm{antenna}}) \cup M_{\mathrm{antenna}}. \tag{22}$$

IV. Proposed NASA-2D/3D View Photos

A. Data Acquisition and Composition

We construct the **NASA-2/3D-Part** dataset based on NASA 3D resources [40], comprising 109 satellite models (e.g., AcrimSat, Aquarius, CloudSat, TDRS). For each model, we provide high-quality triangular meshes with vertex-level annotations for core components, including the main body, solar panels and antennas. In addition to authentic models, we generate augmented samples by disassembling the three components and reassembling them into varied configurations. This augmentation strategy significantly expands the structural diversity of the models, creating a challenging benchmark ideal for evaluating the zero-shot generalization capabilities of segmentation methods.

B. 2D Multi-View Image Rendering

We utilize the Open3D library [34] to perform off-screen rendering and generate multi-view 2D images from the 3D meshes. For each model, we produce a sequence of 200 frames at a resolution of **800×600**. Details are elaborated below.

Viewpoints selection. To ensure comprehensive coverage of the model, we design a continuous spiral camera trajectory around the geometric center of each satellite. The camera completes **three full rotations** (1080°) in azimuth while simultaneously undergoing both radial and vertical motion. Specifically, let $\boldsymbol{C}_{center} = (c_x, c_y, c_z)$ denote the model's centroid and D_{max} the maximum dimension of its bounding box. The camera radius gradually decreases from $1.5 \times D_{max}$ to $0.8 \times D_{max}$, while the elevation linearly increases from $c_z - 0.7 \times D_{max}$ to $c_z + 0.7 \times D_{max}$. This dynamic trajectory captures the target from diverse viewing angles and distances, providing rich parallax and geometric information.

Scene illumination and background. We employ directional lighting to simulate illumination from distant sources such as the Sun, which produces clear shading gradients and shadows that accentuate the geometric features of satellite models. This lighting configuration yields bright and dark sides of the model, enhancing depth perception and structural visibility. The background is set to pure black (RGB: 0, 0, 0) to simulate the dark deep space environment. During rendering, vertex normals are automatically computed for all meshes to ensure proper light interaction with the surface geometry. The camera's up-vector is consistently set to the positive Z-axis to maintain a stable world coordinate orientation across all frames. All rendering operations are performed off-screen at a fixed resolution of 800×600 pixels using Open3D's visualizer.

Camera parameters. For all 200 rendered frames, we meticulously record the complete camera parameters, including the **4×4 extrinsic matrix** defining the camera pose in world coordinates, and the intrinsic parameters encompassing focal length, principal point, and image resolution. These geometric records are essential for downstream tasks such as 3D reconstruction, novel view synthesis, and multi-modal alignment.

C. 3D Point Cloud Sampling & Dataset Organization.

We utilize the Trimesh library [?] to uniformly sample **2048 points** directly from the mesh surface for each model, generating the 3D point cloud representation with preserved vertex-level semantic labels.

To promote standardized evaluation, we implement a deterministic splitting strategy for the 200-frame sequences. Each model's frames are partitioned into a **100-frame training set**, **50-frame validation set**, and **50-frame test set** through fixed random shuffling with model-specific seeds, ensuring reproducibility and consistent splits across all rendering modalities.

V. Experimental results and Analysis

A. Dataset and Evaluation Metrics

The evaluation covers 109 samples. The number of valid images is 5450. The semantic segmentation task targets three distinct parts: body, wing, and antenna. This study employs Intersection over Union (IoU) and Dice Similarity Coefficient (Dice) as evaluation metrics. IoU provides a strict measure of segmentation accuracy by calculating the overlap between predicted and ground-truth regions, being highly sensitive to boundary errors and false positives. In contrast, the Dice coefficient focuses on the similarity between regions, and its heightened sensitivity to overlap offers greater discriminative power for evaluating small targets, such as antennas. The combined use of these two metrics establishes a comprehensive and robust evaluation framework, specifically designed to deliver a precise assessment standard for segmenting small components.

For the class-agnostic detection task, we report the mean Average Precision (mAP) along with AP at IoU thresholds of 50% and 25%. Average Precision is the primary metric for object detection, representing the area under the precision-recall curve and providing a comprehensive measure of a model's accuracy across all recall levels. AP_{50} and AP_{25} correspond to more lenient evaluation criteria, where a detection is considered correct if it exceeds the specified IoU threshold of 0.5 or 0.25 with the ground truth, respectively.

B. Implementation Details

Due to the lack of comparative methods, we evaluate our approach using two distinct metrics: Class-Agnostic Segmentation, which assesses mask assignment accuracy without considering semantic information, and Semantic Segmentation, which evaluates both mask quality and semantic correctness.

1) Class-Agnostic Segmentation: For class-agnostic segmentation, which evaluates mask prediction accuracy independent of semantic labeling, we employ SAM-style automatic mask generation to produce comprehensive image-wide mask proposals. We compare against several SAM-family variants, including SAM, SAM-HQ, MobileSAM, EfficientSAM [41], and SAM2.

2) Semantic Segmentation: For semantic segmentation, we adopt YOLO-World [42] and GroundingDINO [43] as base detectors—both are representative open-domain models in the public research community. Unlike traditional closed-set detectors limited to pre-trained classes, they accept free-text prompts to detect and localize semantically matching objects, enabling flexible adaptation to satellite components without relying on a predefined class list. We then combine them with the SAM family to form comparative methods.

TABLE I
Hyperparameters for Key Modules in Multi-Modal Zero-Shot Segmentation Framework

Convex Isolation	
δ_1	0.7
δ_2	0.3
δ_3	0.5
δ (antenna classification)	0.8
ρ	Adaptive
f	Adaptive
k_1	1
Spatial-guided 3D Semantic Smoothing	
d	pending
Sparse Outlier Filtering	
k	10
β	20%

TABLE II
Performance Metrics Comparison of Different Methods. Higher is better.

Method	Body acc	Antenna acc	Wing acc	All acc	Iou
pointclip	0.4531	0.0811	0.3612	0.4012	34.653
geoze	0.8121	0.1647	0.7068	0.7512	67.416
our	**0.8172**	**0.4731**	**0.7061**	**0.7565**	**68.124**

TABLE III
Class-agnostic mean AP, AP_{50} and AP_{25}. Higher is better.

Method	mean AP	AP_{50}	AP_{25}
SAM2	0.0645	0.0927	0.1501
SAM-HQ	0.0871	0.1255	0.1658
SAM	0.1113	0.1434	0.1784
MobileSAM	<u>0.1236</u>	0.1664	0.2122
EfficientSAM	0.0890	0.1448	0.1852
Ours-SAM2	0.0531	0.0899	0.1555
Ours-MobileSAM	0.0834	0.1450	0.2306
Ours-EfficientSAM	0.0173	0.0366	0.0757
Ours-SAM-HQ	0.1118	<u>0.1878</u>	**0.2735**
Ours-SAM	**0.1353**	**0.2067**	<u>0.2701</u>

All experiments are conducted on a single RTX3090 graphic card. Hyperparameters in our method are summarized in Table I

C. Results and Analysis

The 3D point cloud results are as follows: Our method demonstrates superior performance compared to baseline approaches, as evidenced by the accuracy and IoU metrics reported in Table II. The sparse outlier filtering module effectively mitigates erroneous small-component segmentation by removing outliers, while the antenna preprocessing module elevates the antenna segmentation accuracy to 47.31%

The 2D result analysis is as follows:

1) Class-Agnostic Segmentation: Table III presents the class-agnostic segmentation performance evaluated under the segment-everything protocol. The proposed Ours-SAM achieves the highest mean AP of 0.1353, outperforming the

TABLE IV
SEMANTIC SEGMENTATION RESULTS ON SATELLITE COMPONENTS. METRICS ARE PER-CLASS IOU/DICE AND THEIR MEANS ACROSS CLASSES (MIOU/MDICE); HIGHER IS BETTER.

Method	body IoU	body Dice	wing IoU	wing Dice	ant. IoU	ant. Dice	mIoU	mDice
YOLO-SAM	0.0000	0.0000	0.3189	0.4515	0.0719	0.1222	0.1303	0.1912
YOLO-SAM2	0.0010	0.0019	0.3190	0.4529	0.0685	0.1154	0.1295	0.1901
GD-SAM	0.2554	0.3881	0.4729	0.6154	0.0724	0.1234	0.2669	0.3756
GD-SAM2	0.2430	0.3746	0.4634	0.6042	0.0774	0.1298	0.2612	0.3695
Ours-SAM2	0.3298	0.4818	0.5356	0.6850	**0.0957**	**0.1464**	0.3204	0.4377
Ours-SAM	**0.5691**	**0.7111**	**0.6811**	**0.7941**	0.0878	0.1351	**0.4460**	**0.5467**

strongest baseline MobileSAM. At the stricter IoU threshold of 0.5, Ours-SAM obtains an AP_{50} of 0.2067, demonstrating superior mask quality. Under the more lenient IoU threshold of 0.25, Ours-SAM-HQ achieves the best AP_{25} of 0.2735, indicating effective coverage of target regions.

Qualitative comparisons across multiple methods and view examples are provided in Figure 2, which visually confirms the segmentation quality improvements.

2) Semantic Segmentation: As shown in Table IV, the proposed method demonstrates significant advantages in semantic segmentation of satellite components. Ours-SAM achieves the highest overall performance with a mIoU of 0.4460 and mDice of 0.5467, substantially exceeding the best baseline GD-SAM (mIoU 0.2669, mDice 0.3756).

For major components, Ours-SAM shows remarkable improvements: body IoU reaches 0.5691 compared to 0.2554 by GD-SAM, and wing IoU achieves 0.6811 versus 0.4729. In the challenging antenna category, Ours-SAM2 obtains the best IoU of 0.0957, demonstrating enhanced capability in segmenting thin structures.

The performance ranking consistently places our variants in the top two positions across all metrics. Detector-driven approaches (YOLO-SAM/YOLO-SAM2) exhibit complete failure on the body class, while GroundingDINO-based methods (GD-SAM/GD-SAM2) show limitations in preserving boundaries for fine structures. The proposed framework, by integrating geometry-aware priors with hierarchical fusion, effectively addresses these challenges and produces more consistent segmentation results.

D. Ablation study

To evaluate the effectiveness of different prompt generation strategies in our semantic prompt module, we conduct a comprehensive ablation study comparing three distinct algorithms for semantic prompt point selection. Table V presents the overall accuracy achieved by each algorithm, providing insights into their relative performance and robustness.

TABLE V
ACCURACY COMPARISON OF SEMANTIC PROMPT ALGORITHMS

Prompt Selection Algorithm	Overall Accuracy (%)
kmeans	75.28
hdbscan	69.62
our-dbscan	**78.13**

The comparative analysis reveals significant performance variations among the three prompt selection algorithms. Our-dbscan achieves the highest overall accuracy of 78.13%, demonstrating superior capability in generating semantically meaningful prompt points that effectively guide the segmentation process. Knmeans shows competitive performance with 75.28% accuracy, while hdbscan, with 69.62% accuracy, indicates certain limitations in prompt point selection strategy.

VI. CONCLUSION

In this paper,

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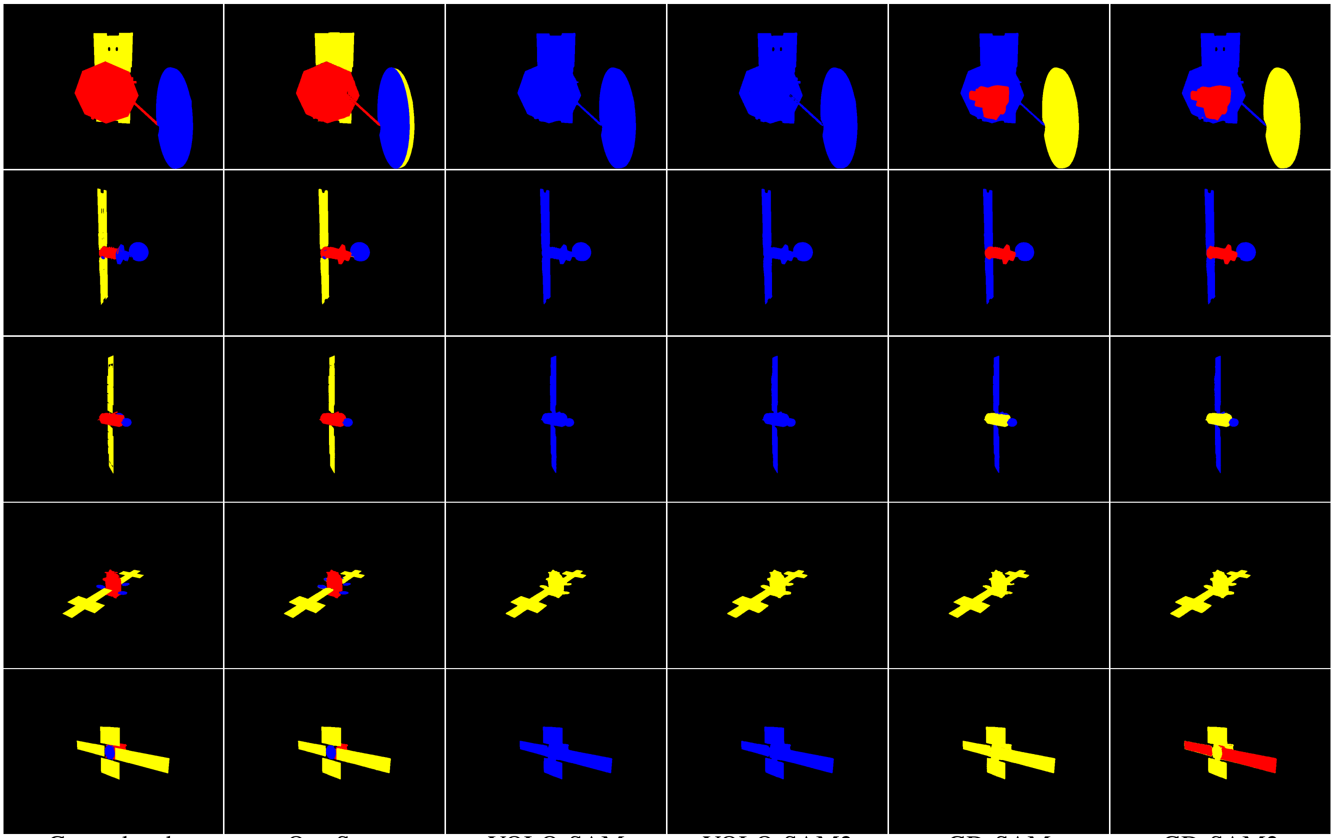

Fig. 2. Qualitative comparisons for semantic segmentation between Our-Sam, YOLO-SAM, YOLO-SAM2, GD-SAM, GD-SAM2.

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